

Dataset Understanding & Forensics

Dissolved Gas Analysis (DGA) — a ground-up study of the three project datasets, from "I know nothing" to research and industrial level.

Project: Intelligent Transformer Health Monitoring using DGA

Datasets: DGA_train.xlsx · DGA_test_unseen.xlsx · IEC_TC_10_data.xlsx

Primary source: IEEE DataPort — "DGA Dataset", DOI 10.21227/27vy-h479

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How to read the evidence tags in this document:

OBSERVED IN FILE measured directly from your Excel files **DOCUMENTATION** stated by the dataset authors
EXTERNAL / VERIFIED from standards / literature **ENGINEERING INTERPRETATION** reasoned conclusion

These four categories are never silently mixed. Where the file and the documentation disagree, both are shown side by side.

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How to use this document

This document exists to move you from "*I have three Excel files I do not understand*" to "*I know exactly what these datasets are, where every number comes from, what each fault means, how an engineer uses this without machine learning, how machine learning uses it, and what the data can and cannot prove.*" It is written for a complete beginner and ramps up to research level. Every technical word is introduced in four steps: the plain idea, the technical name, its engineering meaning, and a concrete example from **your own data**.

It is not a generic DGA article. Almost every claim is tied to one of your three files and tagged with where it came from. A short honesty note before you start: your mentor (me) argued against writing this much theory before you have written any code, because understanding forms fastest while building. You chose the full version, so here it is — but the later chapters on standards, classical methods and machine learning are written at *orientation depth*. They tell you what each thing is and how it connects to your data; you will understand them far more deeply when you build them in code in later steps, and you should expect to revisit those chapters then.

The single most important finding first. Before any theory: I compared the three files row by row. **41 of the 49 rows in the IEC TC 10 file are also present, identical, in the training file (84%).** This means the IEC file cannot be used as an independent "benchmark" after training on the training file — doing so would inflate your scores through data leakage. This one fact changes how you must evaluate your project, and it is explained fully in Part 21. **OBSERVED IN FILE**

Part 1 — What are the three files?

At the simplest level you have three spreadsheets. Each is a separate collection of transformer oil tests. They are not three copies of the same thing; they play three different roles, like a set of practice questions, a sealed exam, and a public reference paper.

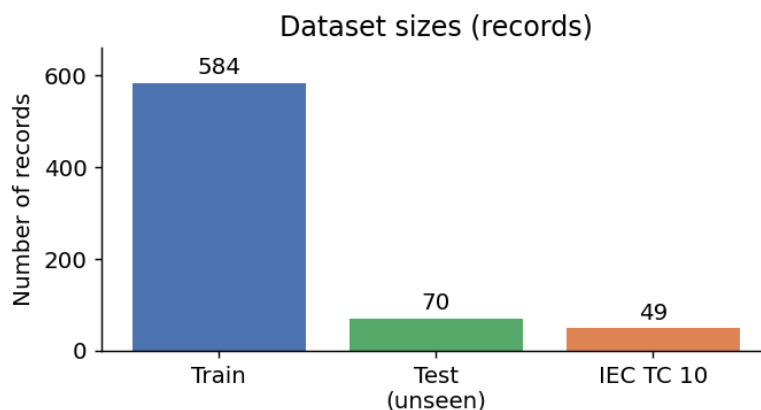


Figure 1. The three files by size. The training file is by far the largest; the IEC file is the smallest.

1.1 DGA_train.xlsx — the teaching file

OBSERVED IN FILE 584 rows, 6 columns, no missing values. **DOCUMENTATION** The authors call it a "class-balanced training dataset ... recommended for training, hyperparameter tuning, and cross-validation."

Plain idea: this is the file the computer studies. **Technical name:** the *training set*. **Meaning:** a model looks at these 584 examples, each showing five gas amounts together with the known correct fault, and it adjusts itself until it can reproduce those answers and, hopefully, generalise to new ones. **Example:** row 1 shows H2=1076, CH4=95, C2H6=4, C2H4=71, C2H2=231 with the answer "PD" — one of 584 such worked examples the model learns from.

Documentation vs file (balance): the documentation says "balanced". Your file is **not** balanced — class D2 has 134 rows (22.9%) while class T2 has only 45 (7.7%), a 3× gap. It is *more even* than the real world, but calling it "balanced" is an overstatement. Full evidence in Part 15. **OBSERVED IN FILE** **DOCUMENTATION**

1.2 DGA_test_unseen.xlsx — the sealed exam

OBSERVED IN FILE 70 rows, exactly 10 per fault class. **DOCUMENTATION** "Unseen test dataset for evaluating model generalization ... use only after finalizing the trained model."

"Test" means you check whether the model truly learned. "Unseen" means the model must not look at these rows while it is being built or tuned — the same reason you do not see the exam paper while revising. If the model is tuned against this file even once, the file stops being a fair exam. **Technical meaning:** it estimates *generalization* — performance on data not used in training.

Note **OBSERVED IN FILE** : this "unseen real-world" file is itself perfectly balanced (10 of every class). Real transformer populations are lopsided, so this file is a *curated* exam, not a picture of real-world fault frequencies. It fairly tests "can the model tell the classes apart," but it does **not** tell you how often each fault occurs in the field.

1.3 IEC_TC_10_data.xlsx — the reference set

OBSERVED IN FILE 49 rows, naturally uneven classes (PD and T1 have just 1 sample each). **DOCUMENTATION** "IEC TC 10 benchmark dataset ... for benchmarking and comparison against an internationally recognized reference dataset."

IEC TC 10 refers to a well-known collection of real transformer fault cases published by CIGRE / used alongside IEC work, which researchers worldwide quote so that different studies can be compared on the same cases. **EXTERNAL / VERIFIED**

What this exact 49-row file is: a small set of those reference cases with the same five gases and the fault labels. Whether it is the complete historical IEC TC 10 collection or a selected subset is **not stated** in the documentation, so treat it as "a reference subset" rather than "the definitive IEC TC 10 set". **ENGINEERING INTERPRETATION**

Its correct role in your project is NOT what the documentation suggests. Because 41 of its 49 rows are duplicated inside the training file (Part 21), you cannot honestly present it as an independent benchmark after training on the training file. Its safe uses: (a) as a source of published worked cases to *validate your classical code* in Step 12, and (b) as a benchmark *only* on the 8 rows that are not in the training file, clearly labelled as such. **OBSERVED IN FILE**

ENGINEERING INTERPRETATION

Part 2 — What is a dataset?

Plain idea: a dataset is a table, exactly like a school attendance register or a shop's sales book. **Technical names:** the table has *rows* (going down) and *columns* (going across); one filled-in box is a *value*; one whole row is an *observation* (also called a *sample* or *record*); a column used as a clue is a *feature*; the column holding the answer is the *target* or *label*.

Mapped onto your data: each **row** is one oil test; the five gas **columns** are the **features** (the clues); the "Fault class" **column** is the **target** (the answer). Here is one real row from the training file translated into plain English:

Row 1 of DGA_train.xlsx: "An oil sample was tested. It contained 1076 units of hydrogen, 95 of methane, 4 of ethane, 71 of ethylene, and 231 of acetylene. The recorded condition was fault type 1 (Partial Discharge)." **OBSERVED IN FILE**

That is the entire machine-learning problem in one line: given the five clues, predict the answer. Everything else in this project is about doing that reliably, and knowing how much to trust the result.

Part 3 — What does one row actually represent?

It is tempting to say "one row = one transformer." That is almost certainly wrong, and the difference matters.

ENGINEERING INTERPRETATION

DOCUMENTATION The instructions state plainly: "Each row represents one DGA sample." A DGA sample is **one oil measurement event** — oil drawn (or measured online) at one moment, giving one set of gas numbers, assigned one condition label. It is a *snapshot of one test*, not a whole transformer and not its life history.

Why this is not the same as "one transformer": a single transformer can be tested many times over years, producing many rows. So one physical transformer could appear several times. **ENGINEERING INTERPRETATION**

What we can and cannot tell from your files **OBSERVED IN FILE**

Question	Answer from the file
Is there a transformer ID column?	No — only 5 gases + label
Is there a timestamp / date?	No
Can we tell if two rows are the same transformer?	No, not reliably — identifying columns were not included
Are there exactly-identical rows?	Yes: 2 exact duplicate rows in Train, 4 rows share an identical gas profile

Why this matters enormously for machine learning. If the same physical transformer's repeated tests landed some rows in Train and some in Test, the model could "recognise" that transformer rather than learn the physics — a hidden form

of cheating (Part 21). Because IDs and dates were removed, you **cannot rule this out**; you can only state it as a limitation. Honest projects say this out loud. **ENGINEERING INTERPRETATION**

Part 4 — What is DGA? (starting before the word)

4.1 What is a transformer, and why does it hold oil?

A **transformer** changes electricity from one voltage to another — for example from the very high voltage used to carry power across the country down to the lower voltage used in a city. Inside are coils of wire (*windings*) wrapped around an iron core. Large power transformers are filled with special **mineral oil**. The oil does two jobs: it **cools** the hot windings by carrying heat away, and it **insulates** — it stops electricity from jumping where it should not. **EXTERNAL / VERIFIED**

4.2 Why problems start, and why gas appears

Over years, two kinds of trouble develop. **Too much heat** (an overloaded or degrading spot) and **electrical discharge** (electricity sparking through a weak point). Both do the same underlying thing: they deliver energy to the oil and the paper insulation, and that energy **breaks the molecules apart**. Oil is made of hydrogen and carbon atoms joined together; when those joins break, the freed atoms recombine into small gas molecules — hydrogen, methane, ethylene, acetylene and others. These gases **dissolve into the oil** and stay there, like fizz dissolved in a soft drink. **EXTERNAL / VERIFIED**

4.3 Dissolved • Gas • Analysis

Dissolved = the gases are mixed into the liquid oil, not floating above it. **Gas** = the small molecules created by heat or sparking. **Analysis** = measuring how much of each gas is present. Put together, **Dissolved Gas Analysis** is: take a little oil, measure the gases hiding in it, and read those amounts as clues to what is happening inside a sealed transformer you cannot open.

Analogy: the oil is like the transformer's blood, and DGA is like a blood test — you draw a small sample and infer the health of the whole body from what is dissolved in it. **Where the analogy breaks:** blood tests measure things the body needs; here the gases are *waste products of damage* that should ideally be near zero. And unlike blood, a specific *mix* of gases points to a specific *kind* of fault — that precision is the whole point.

Part 5 — Why does DGA work? (the physics and chemistry)

The reason DGA is useful is that **different faults break the oil in different ways, producing different gas mixtures**. This is the central idea, so it is worth building slowly. **EXTERNAL / VERIFIED**

How much energy / temperature	What tends to break	Gases that dominate
Gentle, low-energy sparking (partial discharge)	weakest bonds only	mostly H_2 (hydrogen)
Moderate overheating (a few hundred °C)	medium bonds	CH_4 , then C_2H_4 as it gets hotter
Very hot faults (>700 °C)	strong bonds	lots of C_2H_4 (ethylene)
Violent high-energy arcing (like a lightning strike inside)	everything, extremely fast	C_2H_2 (acetylene) — needs the most energy

So the pattern of gases is a **fingerprint**. A fingerprint rich in acetylene screams "violent arcing"; one that is almost only hydrogen whispers "gentle partial discharge"; one heavy in ethylene says "something is running very hot." **EXTERNAL / VERIFIED** Your own data shows this fingerprint effect directly — see Figure 6 in Part 13.

Part 6 — Understanding every gas

Your files contain five gases. For each, here is the beginner meaning, why it appears, what it signals, and its actual range in **your training file**. **EXTERNAL / VERIFIED** **OBSERVED IN FILE**

H_2 — Hydrogen

the smallest, easiest molecule to make. It is the "first responder" gas — almost every fault makes some. On its own, high hydrogen with little else suggests **partial discharge** (gentle sparking). It rarely means much alone; it needs the company of the other gases to be read properly. **In your training file:** minimum 0.0000, median 78.0, mean 1254.2, maximum 92600 ppm. **OBSERVED IN FILE**

CH₄ — Methane

forms from mild-to-moderate heating. Rising methane alongside hydrogen points toward **thermal** (heat) problems rather than pure sparking. **In your training file:** minimum 0.0000, median 45.2, mean 684.1, maximum 61000 ppm.

OBSERVED IN FILE

C₂H₆ — Ethane

a marker of **lower-temperature heating**. Ethane rising while ethylene stays low suggests a warm, not hot, fault. **In your training file:** minimum 0.0000, median 25.0, mean 232.4, maximum 29990 ppm.

OBSERVED IN FILE

C₂H₄ — Ethylene

the signature of **hot thermal faults**. High ethylene means high temperature — the hotter it gets, the more ethylene relative to ethane. **In your training file:** minimum 0.0000, median 31.9, mean 725.2, maximum 42100 ppm.

OBSERVED IN FILE

C₂H₂ — Acetylene

the "alarm" gas. It takes the most energy to form, so meaningful acetylene almost always means **high-energy discharge / arcing (D2)** — the most urgent fault. In your training file 77 rows have exactly zero acetylene and 176 more use the 0.0001 placeholder, i.e. a large share of samples have essentially none — which is exactly what you expect, since most faults are not arcing. **In your training file:** minimum 0.0000, median 2.5, mean 486.5, maximum 57000 ppm.

OBSERVED IN FILE

The golden rule for all five: never read one gas alone. A single gas is a letter; the fault name is spelled by the whole word. This is exactly why the classical methods (Part 27) and the machine-learning model (Part 30) look at combinations and ratios, not single numbers.

EXTERNAL / VERIFIED

Part 7 — What is ppm?

ppm means **parts per million**. Imagine a box of one million identical grains of sand. If 5 of those grains are red, the red concentration is "5 ppm". It is simply *how many out of a million*. For gases in oil it means microlitres of gas per litre of oil — a tiny fraction.

EXTERNAL / VERIFIED

Two beginner traps ppm helps you avoid: (1) a "big" number is not automatically alarming — hydrogen in the hundreds can be normal, while acetylene at 20 can be serious, because each gas has its own meaningful scale; (2) the **numbers span a huge range**. In your training file hydrogen runs from 0 to about 92600 ppm and acetylene from 0 to about 57000 ppm.

OBSERVED IN FILE

. That is why later steps often use ratios or logarithms — to compare gases fairly rather than let the biggest raw number dominate.

Part 8 — Understanding the seven fault labels

DOCUMENTATION The dataset uses an integer code 0–6. Do not stop at the mapping — here is what each physically means.

EXTERNAL / VERIFIED

Code	Short	Family	What is physically happening	Gas signature
0	Normal	—	No fault detected by DGA (see Part 9 — this is not the same as "perfect")	all gases low
1	PD	Electrical	Partial discharge: tiny sparks in gas bubbles / weak insulation spots — gentle but a warning sign	mostly H ₂
2	D1	Electrical	Low-energy discharge: sparking / tracking with modest energy	H ₂ , C ₂ H ₂ present
3	D2	Electrical	High-energy discharge / arcing: a powerful arc — the most dangerous, fast damage	strong C ₂ H ₂
4	T1	Thermal	Thermal fault < 300 °C: mild overheating	CH ₄ , C ₂ H ₆
5	T2	Thermal	Thermal fault 300–700 °C: moderate overheating	C ₂ H ₄ rising
6	T3	Thermal	Thermal fault > 700 °C: severe overheating	strong C ₂ H ₄

The three families to remember: electrical faults involving sparking (PD, D1, D2), thermal faults involving heat (T1, T2, T3), and Normal. Electrical severity climbs PD → D1 → D2; thermal severity climbs T1 → T2 → T3 with temperature.

EXTERNAL / VERIFIED

Part 9 — "Normal" does not mean "perfectly healthy"

This distinction is subtle and important. **ENGINEERING INTERPRETATION** "Normal" (class 0) means **no fault was detected by dissolved-gas analysis**. It does **not** guarantee the transformer is flawless. A fault may be too early to have produced much gas yet; or a problem may be of a type DGA does not see (for example purely mechanical looseness).

A useful ladder of meaning, from safest to most certain-bad: **normal** (no gas evidence) → **no detected abnormality** → **early degradation** (gases creeping up) → **developing fault** → **confirmed fault**. Your label only distinguishes "normal" from named faults; it does not capture the in-between. So the honest statement is "DGA shows no fault," never "the transformer is certified healthy." Note also (Part 27) that some classical tools, such as the Duval Triangle, **cannot output "normal" at all** — they always name a fault — which is another reason "normal" must be handled carefully.

Part 10 — The column dictionary (master reference)

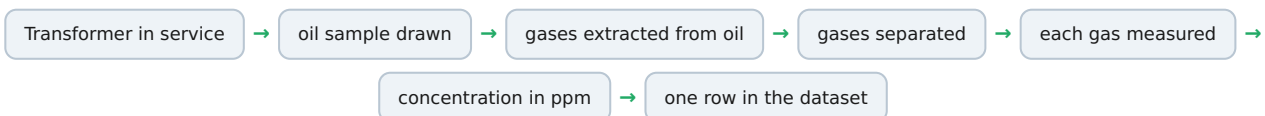
Every column in all three files, in plain English. **OBSERVED IN FILE** **EXTERNAL / VERIFIED**

Column	Meaning	Unit	Physical meaning	Diagnostic meaning	Role in ML
H ₂	Hydrogen	ppm	smallest gas, made by almost any fault	sparking / partial discharge marker	input feature
CH ₄	Methane	ppm	from mild heating	low-temp thermal marker	input feature
C ₂ H ₆	Ethane	ppm	from low-temperature heating	warm thermal marker	input feature
C ₂ H ₄	Ethylene	ppm	from hot heating	high-temp thermal marker	input feature
C ₂ H ₂	Acetylene	ppm	needs the most energy	arcing / high-energy discharge alarm	input feature
Fault class	Condition code 0-6	integer	the diagnosed state	the answer being predicted	target / label

Units caveat **DOCUMENTATION** : the instructions say the gas columns are "dissolved gas concentration values (as provided)" and do not print the unit explicitly. DGA concentrations are conventionally ppm, and the value ranges are consistent with ppm, so this document treats them as ppm — but strictly, the unit is *assumed by convention, not stated by the source*. **ENGINEERING INTERPRETATION**

Part 11 — How the gas values are actually measured

The numbers in your spreadsheet are the last link in a physical chain. **EXTERNAL / VERIFIED**



Gas Chromatography (GC) — the laboratory method

The classic way. The dissolved gases are pulled out of the oil and pushed through a long thin tube packed with material that different gases travel through at different speeds. They arrive one at a time and a detector measures each — giving the ppm of hydrogen, methane, and so on. This is the traditional laboratory DGA method and the most likely origin of reference data like these. **EXTERNAL / VERIFIED**

Photoacoustic Spectroscopy (PAS) — the online method

A newer approach used in sensors bolted to a transformer for continuous monitoring: a gas absorbs a pulse of light, warms slightly, and makes a tiny sound that a microphone measures; the sound size gives the concentration. It matters for real-time monitoring, but there is **no evidence your Excel files came from PAS** — the source does not say, so do not claim it.

ENGINEERING INTERPRETATION

Part 12 — Where the data comes from (provenance)

Being honest about origins is part of the science. Here is what is confirmed, likely, and simply not documented.

DOCUMENTATION

ENGINEERING INTERPRETATION

Item	Status	What we know
Hosting / primary source	Confirmed	IEEE DataPort, "DGA Dataset", DOI 10.21227/27vy-h479, formats .csv/.xlsx
Variables	Confirmed	5 gases (H_2 , CH_4 , C_2H_6 , C_2H_4 , C_2H_2) + integer fault class 0–6
IEC TC 10 lineage	Probable	the 49-row file corresponds to the internationally used IEC TC 10 / CIGRE reference fault cases
Original transformers (make, age, voltage, country)	Not documented	no such columns; no population description given
Measurement method (GC vs PAS)	Not documented	not stated for these files
How labels were assigned	Not documented	whether by inspection, by standards, or by expert opinion is not stated
Collection dates / timeline	Not documented	no timestamps present

The large number of "not documented" rows is not a criticism of the dataset — small public research datasets are usually anonymised. But it directly limits what you may claim (Part 35). You can say "this method classifies these documented cases well"; you cannot say "this proves performance on Indian 220 kV transformers," because you do not know if any are in the data.

ENGINEERING INTERPRETATION

Part 13 — The actual numbers in your files

Everything here is measured directly from your three Excel files.

OBSERVED IN FILE

13.1 Gas statistics — Training file

Gas	min	25%	median	mean	75%	max	std
H_2 (Hydrogen)	0.0	17.0	78.0	1,254.2	279.2	92,600.0	6,485.1
CH_4 (Methane)	0.0	13.8	45.2	684.1	171.0	61,000.0	3,310.1
C_2H_6 (Ethane)	0.0	3.9	25.0	232.4	92.2	29,990.0	1,710.0
C_2H_4 (Ethylene)	0.0	4.0	31.9	725.2	197.2	42,100.0	3,281.6
C_2H_2 (Acetylene)	0.0	0.0	2.5	486.5	71.2	57,000.0	3,107.0

13.2 Gas statistics — Test (unseen) file

Gas	min	25%	median	mean	75%	max	std
H_2 (Hydrogen)	0.0	20.0	84.8	1,089.7	212.8	33,046.0	4,274.8
CH_4 (Methane)	0.0	13.6	49.8	437.7	223.2	17,424.0	2,085.4
C_2H_6 (Ethane)	0.0	5.1	22.0	169.5	88.2	7,299.0	869.3
C_2H_4 (Ethylene)	0.0	5.6	37.2	704.9	143.0	37,043.0	4,420.4
C_2H_2 (Acetylene)	0.0	0.0	1.1	91.3	29.1	1,827.0	310.1

13.3 Gas statistics — IEC TC 10 file

Gas	min	25%	median	mean	75%	max	std
H ₂ (Hydrogen)	0.2	21.2	400.0	1,536.8	1,330.0	20,000.0	3,307.1
CH ₄ (Methane)	0.0	9.7	89.0	2,614.3	966.0	64,064.0	9,440.0
C ₂ H ₆ (Ethane)	0.0	0.2	20.0	2,332.5	128.0	72,128.0	11,055.3
C ₂ H ₄ (Ethylene)	0.0	0.4	153.0	4,230.5	820.0	95,650.0	14,700.0
C ₂ H ₂ (Acetylene)	0.0	0.0	26.0	1,494.4	325.0	57,000.0	8,122.5

Two things jump out. First, in every file the **mean is far larger than the median** for each gas (e.g. training hydrogen: median 78 but mean 1254). That gap is the mathematical signature of a **right-skewed** distribution — most samples are small, a few are enormous. Second, the **IEC file reaches much higher values** (ethylene up to 95650 ppm) than the others, consistent with it holding severe real fault cases. **OBSERVED IN FILE**

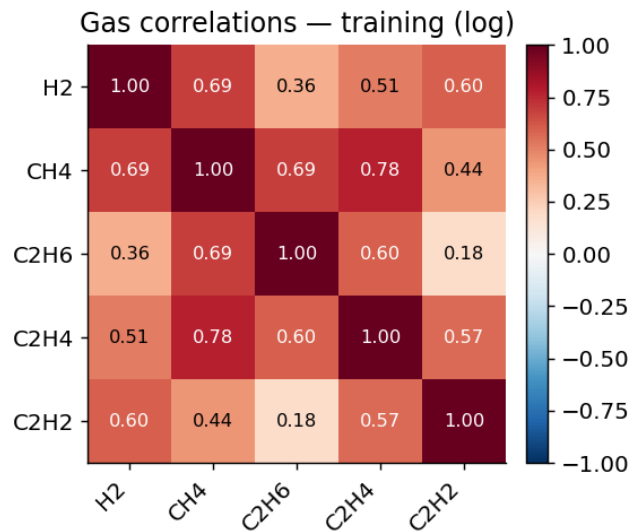


Figure 2. How the gases move together in the training file (on a log scale). Values near +1 mean two gases tend to rise together. Thermal gases (methane, ethane, ethylene) are related; acetylene is more its own signal.

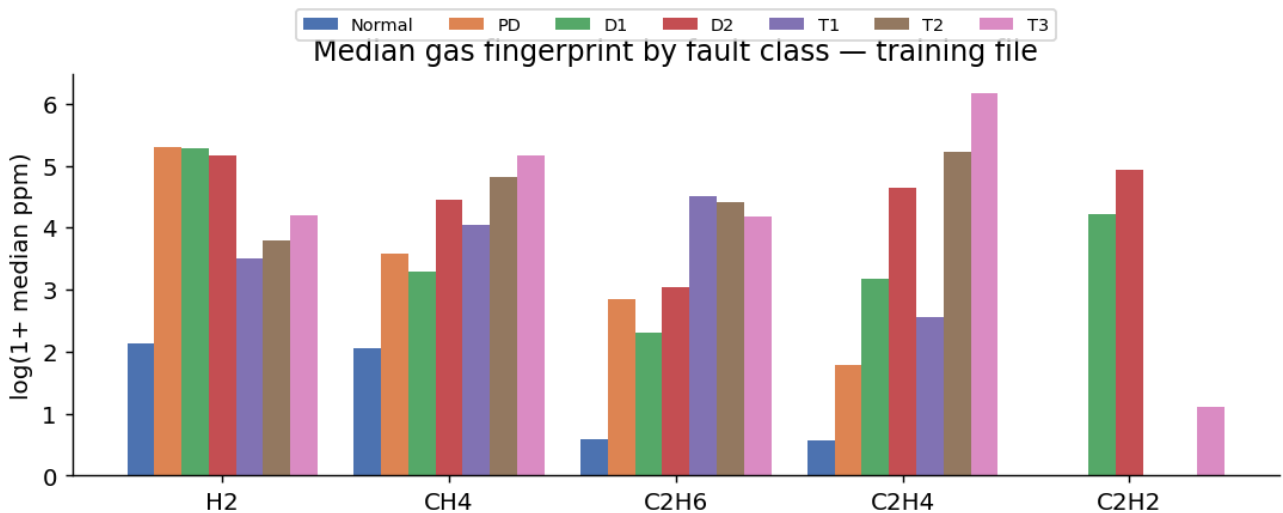


Figure 3. The median "fingerprint" of each fault class in the training file. Note how the arcing class (D2) stands out on acetylene, and the hot thermal class (T3) on ethylene — the physics of Part 5 showing up in your own data.

Part 14 — Class distribution across the three files

Fault class	Train — n (%)	Test (unseen) — n (%)	IEC TC 10 — n (%)
0 — Normal	85 (14.6%)	10 (14.3%)	15 (30.6%)
1 — PD	59 (10.1%)	10 (14.3%)	1 (2.0%)
2 — D1	76 (13.0%)	10 (14.3%)	7 (14.3%)
3 — D2	134 (22.9%)	10 (14.3%)	15 (30.6%)
4 — T1	96 (16.4%)	10 (14.3%)	1 (2.0%)
5 — T2	45 (7.7%)	10 (14.3%)	2 (4.1%)
6 — T3	89 (15.2%)	10 (14.3%)	8 (16.3%)
Total	584	70	49

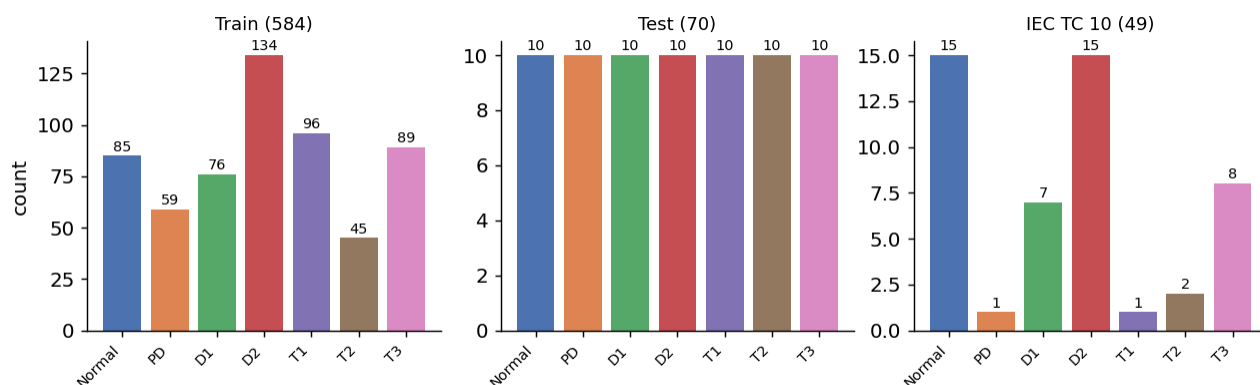


Figure 4. Fault-class counts in each file. Train is uneven, Test is flat (10 each), IEC is steeply uneven.

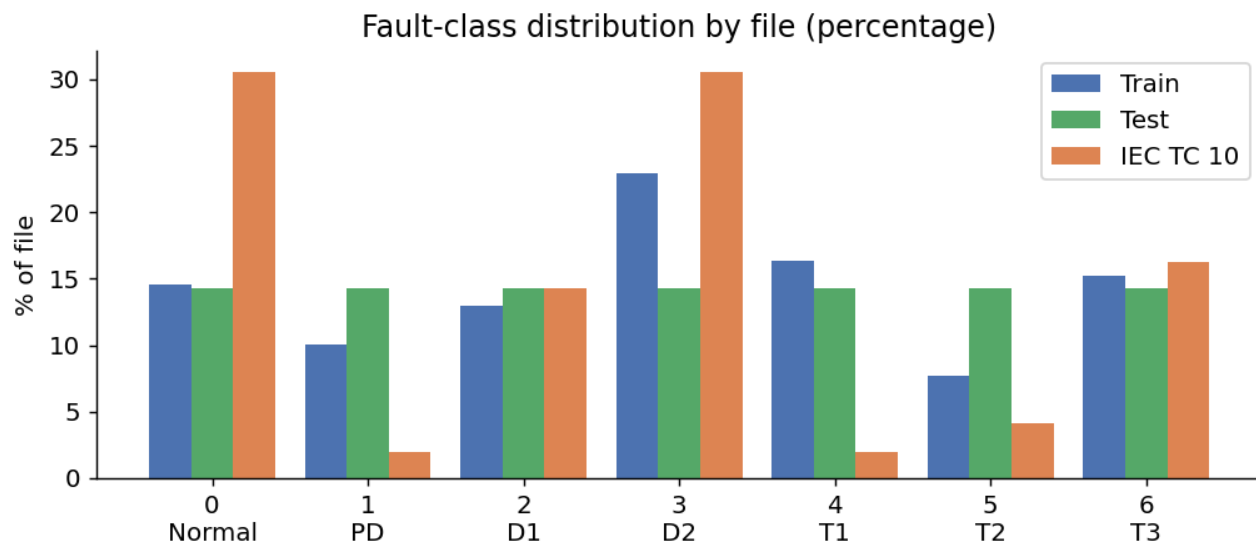


Figure 5. The same information as percentages, so the three differently-sized files can be compared directly.

Reading it: the largest training class is D2 (arcing) at 134; the smallest is T2 at 45. In the IEC file, PD and T1 have a **single** sample each — you cannot meaningfully learn or fairly test a class from one example. **OBSERVED IN FILE** Why this matters is Part 24 (imbalance): a model can look accurate overall while completely failing a rare-but-dangerous class.

Part 15 — Documentation vs file: is the training set really "balanced"?

The documentation makes a specific claim. Your file gives a specific answer. Here they are side by side.

DOCUMENTATION

OBSERVED IN FILE

Claim in documentation	What the actual file shows	Interpretation	What to verify
"Balanced training dataset"	classes range 45 (T2, 7.7%) to 134 (D2, 22.9%) — a 3× spread	Partly true: far more even than the real world, but not balanced in the strict sense of equal counts	Whether "balanced" meant "resampled toward balance" rather than "equal". The source does not say the method.
"Unseen real-world testing dataset"	exactly 10 per class — perfectly flat	"Unseen" is plausible; "real-world" is questionable, since real fault frequencies are never flat	Whether the test rows are genuinely independent transformers (cannot confirm — no IDs)
IEC = "benchmark ... internationally recognized reference"	41 of 49 rows are duplicated inside Train	Cannot serve as an independent benchmark after training on Train (leakage)	Use only the 8 non-overlapping rows as a benchmark, or de-duplicate first

Rule you just practised: when the documentation and the file disagree, you do not silently trust either — you report both and say what remains to be verified. That habit is worth more to an interviewer than any accuracy score.

ENGINEERING INTERPRETATION

Part 16 — The shape of the gas distributions

Gas value distributions — training file (log scale)

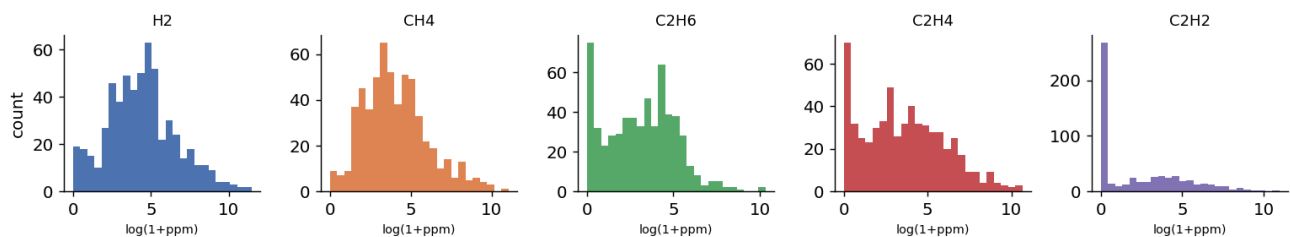


Figure 6. How often each value occurs for each gas (training file, log scale). The tall left side means most samples have small amounts; the long tail means a few have very large amounts.

Spread of each gas — training file (log scale)

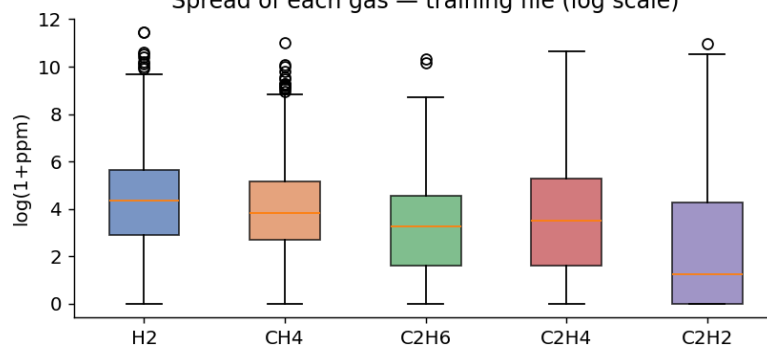


Figure 7. Box plots of the same five gases. The box holds the middle half of the data; the line inside is the median; the dots are unusually large values (outliers).

Plain reading: every gas is **right-skewed** — a big pile of small values and a thin tail of very large ones. **Why it matters:** a few huge samples can dominate any calculation that uses raw sizes, and can dominate what a model pays attention to. This is the practical reason later steps use **ratios** (Part 25) and often a **logarithm** (which pulls the long tail back in) before modelling.

OBSERVED IN FILE

ENGINEERING INTERPRETATION

Part 17 — The zero and near-zero values

Your files contain both true zeros and a placeholder value of exactly 0.0001. Here is the count of each. **OBSERVED IN FILE**

Gas	true zeros (Train)	0.0001 placeholder (Train)	true zeros (Test)	0.0001 (Test)	true zeros (IEC)
H ₂	1	15	0	3	0
CH ₄	2	6	0	1	2
C ₂ H ₆	13	33	0	4	8
C ₂ H ₄	7	27	0	3	5
C ₂ H ₂	77	176	5	23	15

What a zero means here: most likely "the gas was below what the instrument can detect," i.e. essentially none — which is normal and expected, especially for acetylene, since most samples are not arcing. **ENGINEERING INTERPRETATION** The 0.0001 values look like a deliberate substitute for zero so that later **ratio** calculations (Part 25) do not divide by zero and crash.

Important honesty point: the documentation does **not** explain the 0.0001 values, and the substitution is **inconsistent** — the same column contains both true 0 and 0.0001. So "0.0001 = safe substitute for zero" is a well-supported *interpretation*, not a confirmed fact. When you write ratio code, handle both cases explicitly, and note this in your decision log.

ENGINEERING INTERPRETATION

Part 18 — Missing data

A **missing value** is an empty cell — a measurement that was never recorded. They matter because most models cannot accept a blank and because a *pattern* of missingness can bias results. **EXTERNAL / VERIFIED**

Good news, measured directly: all three of your files have **zero missing values** in every column. **OBSERVED IN FILE** You will not need imputation (filling gaps) for this data. Do not confuse this with the zeros of Part 17 — a zero is a recorded measurement of "almost none"; a missing value would be no measurement at all, and you have none of those.

Part 19 — Outliers (unusually large values)

An **outlier** is a value far larger (or smaller) than the rest. In many datasets an outlier is an error to remove. In DGA, an outlier is often the **most important row in the file** — a genuinely severe fault with genuinely huge gas numbers.

EXTERNAL / VERIFIED

Real example from your training file: the row with the most acetylene has C₂H₂ = 57000 ppm, with fault class 3. That is not noise to delete — it is a textbook arcing case. **OBSERVED IN FILE**

Rule: do not automatically delete large values in DGA. Decide with engineering context: an impossible value (e.g. negative, or larger than physically possible) may be an error; a very large but plausible value is probably a real severe fault and is exactly what your system must catch. **ENGINEERING INTERPRETATION**

Part 20 — Duplicates and repeated observations

Measured directly **OBSERVED IN FILE** : the training file has **2 exactly-identical rows** and **4 rows that share an identical gas profile** (some with the same label, some possibly differing). The test file has 0 exact duplicates; the IEC file has 1 repeated gas profile.

Why it matters: exact duplicates inside the training file let a model "memorise" a repeated case and can mildly inflate cross-validation scores. Identical gas values carrying *different* labels are worse — they set a hard ceiling on achievable accuracy, because no model can separate two rows that look identical but are labelled differently. And duplicates that cross *between* files become leakage, which is Part 21. **ENGINEERING INTERPRETATION**

Part 21 — Data leakage (the critical finding)

What it is, plainly: data leakage is when the exam questions were secretly included in your revision material. You "score" brilliantly, but the score is a lie — you were tested on what you had already seen. **Technical meaning:** information from the evaluation set leaks into training, so the measured performance overstates real-world performance. **EXTERNAL / VERIFIED**

21.1 What I found in your files

File pair	Identical full rows shared	Identical gas profiles shared
Train ↔ Test (unseen)	0	1
Train ↔ IEC TC 10	41 of 49	43
Test ↔ IEC TC 10	0	0

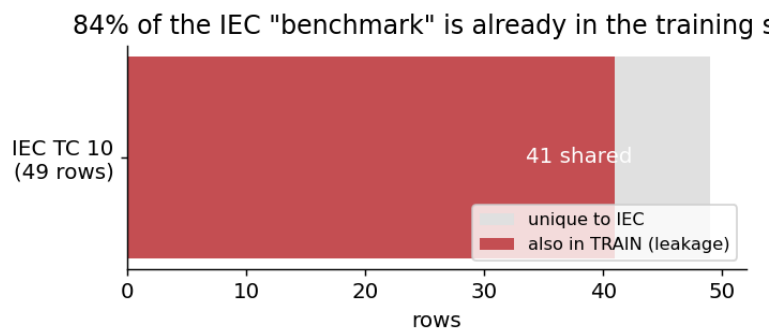


Figure 8. 41 of the 49 IEC rows are already in the training file.

21.2 What it means for your project

The good news: your Train and Test files are essentially clean of each other (0 shared full rows). So the standard "train on Train, judge on Test" workflow is safe from leakage between those two.

The problem: the documentation tells you to "report benchmark results on IEC TC 10." If you train on Train and then score on IEC, **84% of that IEC "benchmark" was in your training data** — the score would be inflated and scientifically worthless as an independent benchmark.

What to do (record this in DECISIONS.md): either (a) benchmark only on the 8 IEC rows that are NOT in Train, clearly labelled as a tiny sample; or (b) remove the 41 overlapping rows from Train before any IEC benchmarking; and (c) never quote an IEC score without stating how you handled this overlap. **OBSERVED IN FILE** **ENGINEERING INTERPRETATION**

One caveat, stated honestly: because transformer IDs were removed (Part 3), there could **also** be a subtler leakage — the same physical transformer appearing in both Train and Test under different rows. You cannot detect this from the data, so you note it as an unverifiable limitation rather than a proven problem. **ENGINEERING INTERPRETATION**

Part 22 — Is there time information?

Measured directly **OBSERVED IN FILE** : there is **no date, time, or sequence column** in any file. Each row is a standalone snapshot with no "before/after" relationship to any other.

Consequence — be precise in your claims. **ENGINEERING INTERPRETATION** This is a **snapshot** dataset, not a **time-series** dataset. Therefore it can legitimately support *classifying the state of a sample* (what fault is this?). It **cannot** support genuine *trending*, *rate-of-gas-generation*, real-time monitoring claims, or **remaining useful life** predictions — those all need repeated measurements of the same transformer over time, which this data does not contain. Promising RUL from this data would be a red flag to any reviewer.

Part 23 — Information we have vs information we do NOT have

We HAVE OBSERVED IN FILE	We DO NOT have (not in the files)
5 gas concentrations per sample	transformer ID, age, manufacturer
an integer fault label 0–6	MVA rating, voltage class, loading
584 / 70 / 49 samples in three roles	operating temperature, oil type, moisture, oxygen, nitrogen
clean data (no missing values)	location, maintenance history, previous DGA, timestamps

Why the missing context limits conclusions: without loading, moisture, or age you cannot separate "this transformer is faulty" from "this transformer is simply old / heavily loaded," and you cannot correct for conditions that raise gas levels harmlessly. Your model learns "gas pattern → fault label" and nothing about the machine's circumstances. That is a valid problem to solve — but it is a narrower claim than "full transformer health assessment," and you should describe it that way.

ENGINEERING INTERPRETATION

Part 24 — What is class imbalance, and why accuracy can lie?

Balanced = each class has roughly equal numbers. **Imbalanced** = some classes (the *majority*) are common and others (the *minority* or *rare* classes) are scarce. **Classroom analogy:** if 95 students passed and 5 failed, a lazy teacher who marks *everyone* "pass" is right 95% of the time — while completely failing to spot the 5 who need help. **EXTERNAL / VERIFIED**

Numerical example with the danger spelled out: suppose arcing (D2) is 5% of real samples. A model that *always* predicts "not D2" scores 95% accuracy while catching **zero** arcing faults — the single most dangerous condition. That is why **overall accuracy is a dangerous headline number** for this project, and why you will report **per-class** results (precision and recall for each fault) instead. Your IEC file, with 1 sample each of PD and T1, is exactly where this bites.

ENGINEERING INTERPRETATION

From here the document shifts to orientation depth. Parts 25–36 explain the engineering and machine-learning ideas your project will use, and how they connect to this data. You will *build* these in later steps (9–21) and understand them far more deeply with code in front of you. Read these as a map, not a manual. **ENGINEERING INTERPRETATION**

Part 25 — Raw values vs engineered features (ratios)

A **raw feature** is a number straight from the file (e.g. $C_2H_2 = 231$). A **derived / engineered feature** is a new number you compute from the raw ones because it carries the meaning more directly. In DGA the classic engineered features are **gas ratios**. **EXTERNAL / VERIFIED**

Why ratios: the *proportion* of one gas to another reflects the energy of the fault better than the raw amount, and it partly cancels out "how much oil / how big the transformer" effects. The IEC 60599 method uses three: C_2H_2/C_2H_4 , CH_4/H_2 , and C_2H_4/C_2H_6 .

Ratio	Top / bottom	Roughly indicates
C_2H_2 / C_2H_4	acetylene over ethylene	arcing vs pure heat
CH_4 / H_2	methane over hydrogen	thermal vs electrical tendency
C_2H_4 / C_2H_6	ethylene over ethane	how hot the thermal fault is

Direct link to Part 17: the bottom of a ratio can be zero (e.g. no hydrogen). Dividing by zero crashes code and, at very low concentrations, a tiny measurement wobble swings a ratio wildly, making it noise. This is why the 0.0001 placeholders exist and why your ratio code must guard against a zero or near-zero denominator and be allowed to return "no decision."

ENGINEERING INTERPRETATION

A related visual tool, the **Duval Triangle**, turns three gases (CH_4 , C_2H_4 , C_2H_2) into a single point on a triangle whose zones name the fault. Your five gases support the Duval Triangle; the Duval Pentagon needs two more gases you do not have, so it is only partly applicable. You will build the triangle in Step 11. **EXTERNAL / VERIFIED**

Part 26 — What are international standards?

Plain idea: a standard is an agreed rulebook so everyone does something the same way — like everyone agreeing which side of the road to drive on. In engineering, standards let a lab in India and a lab in Germany interpret the same oil test the same way. **EXTERNAL / VERIFIED**

Body / standard	What it is	Role in your project
IEEE	Institute of Electrical and Electronics Engineers — a global professional body that publishes standards	publisher of C57 series below; also hosts your dataset
IEC	International Electrotechnical Commission — international standards for electrical technology	publisher of 60599 (ratio method) and the TC 10 reference cases
IEEE C57.104	guide for interpreting DGA in oil-immersed transformers (gas limits / status levels)	defines "how much gas is too much" — used to decide something is abnormal before naming it
IEC 60599	the international ratio-method standard	the boundaries your IEC-ratio code (Step 10) will implement
IEEE C57.91	loading guide for oil-immersed transformers	background on how loading/temperature stress relates to ageing

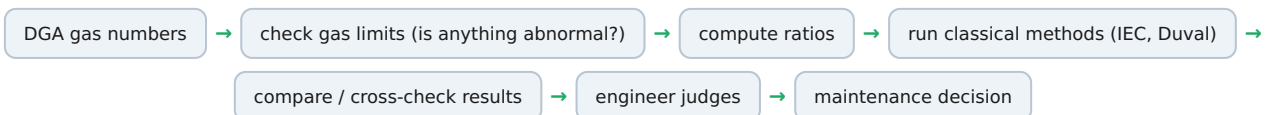
Part 27 — Classical diagnostic methods (no machine learning)

A **classical method** is a fixed human-made rule for reading gases — no learning from data, just "if the gases look like this, call it that." They are decades old and trusted. **EXTERNAL / VERIFIED**

Method	Idea	Output	Main weakness
Key Gas	which single gas dominates?	a fault family	crude; ignores combinations
Doernenburg ratios	ratios with validity checks	fault type or "not applicable"	needs gases above threshold
Rogers ratios	ratio bands → fault codes	fault type	gaps where no code fits
IEC 60599 ratios	three ratios vs standard boundaries	fault type or "no decision"	fails at very low gas levels
Duval Triangle	three gases → a point in a zone	always names a fault	can never say "normal"

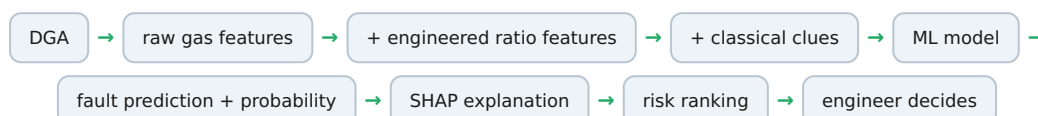
Two traps to carry forward. (1) The "no decision" problem: honest ratio methods sometimes refuse to answer — and your code must be allowed to do the same rather than force a guess. (2) The "healthy" problem: the Duval Triangle *always* outputs a fault, so you must first use gas limits (IEEE C57.104) to decide whether anything is wrong at all, and only then use the triangle to name it. **EXTERNAL / VERIFIED**

Part 28 — How an engineer diagnoses WITHOUT machine learning



Worked on a real row of yours (training row 1: $H_2=1076$, $CH_4=95$, $C_2H_6=4$, $C_2H_4=71$, $C_2H_2=231$): hydrogen is high and acetylene is present — not a purely thermal picture. The ratios and Duval position would push toward a discharge-type fault; the recorded label is PD (class 1). An engineer would weigh the numbers, note any borderline ratios, and decide whether to monitor more closely or intervene. **OBSERVED IN FILE** **ENGINEERING INTERPRETATION**

Part 29 — What changes when we ADD machine learning?



What ML adds: it can weigh all five gases and their ratios together, learn subtle combinations no single rule captures, output a **probability** ("78% D2") rather than a bare label, and rank a whole fleet by risk. **What ML does NOT replace:** the standards that decide "is this abnormal," the physics that makes the features meaningful, and the engineer who owns the final decision. Your project deliberately builds the classical layer *first* so the ML has something proven to beat and to lean on.

ENGINEERING INTERPRETATION

Part 30 — What exactly does the model learn?

Plain version: the model sees hundreds of examples of "these gas amounts went with this fault" and gradually works out statistical patterns linking the two, so it can guess the fault for new gas amounts. **Technical version:** it learns a function from the five-dimensional gas space to a probability over seven classes, by minimising its error on the training examples.

EXTERNAL / VERIFIED

The distinction that keeps you honest: the model does **not** understand transformers the way an engineer does. It has never seen a transformer; it only knows the patterns in the 584 rows it was shown. So it inherits every quirk and limit of this data — the imbalance, the anonymisation, the curated balance. That is precisely why understanding the dataset (this whole document) comes before trusting any model built on it.

ENGINEERING INTERPRETATION

Part 31 — Why the dataset size matters

584 training rows is **small** by machine-learning standards (modern deep learning often uses millions). **EXTERNAL / VERIFIED** Small data has a specific danger: **overfitting** — a model powerful enough can *memorise* the training rows, scoring perfectly there and failing on anything new, like a student who memorises answers without understanding.

This is why deep neural networks are the **wrong tool here**, and why classical machine-learning models that do well on small tables are the right candidates: **Support Vector Machine (SVM)**, **Random Forest**, **XGBoost**, **LightGBM**. You are not training them yet — just know these are the sensible options and why.

ENGINEERING INTERPRETATION

Part 32 — Training, validation, testing — mapped to your files

Concept	Plain meaning	Your file
Training	examples the model studies	DGA_train.xlsx
Validation	a slice held out <i>from training</i> to tune choices, via cross-validation	carved out of DGA_train.xlsx (e.g. stratified k-fold)
Test (unseen)	the sealed final exam, used once	DGA_test_unseen.xlsx
External benchmark	an outside reference for comparison	IEC TC 10 — with the leakage caveat of Part 21

Do not blindly follow the documentation's workflow. "Train → test on unseen" is sound. But "report benchmark on IEC" needs the Part 21 fix first. Because class T2 has few samples, your validation split must be **stratified** (each fold keeps the class proportions) so a rare class does not vanish into one fold. Record this choice in DECISIONS.md.

ENGINEERING INTERPRETATION

Part 33 — What can the IEC dataset legitimately tell us?

An external reference is valuable because doing well on your *own* test proves less than doing well on data assembled by *someone else*. If performance holds up on a true external set, that is real evidence of generalisation; if it drops sharply, it warns that your model learned quirks of the training source.

EXTERNAL / VERIFIED

For your specific IEC file, that logic only holds for the **8 rows not already in Train**. On the 41 overlapping rows, a strong score proves nothing (the model saw them). So the honest framing is: "IEC used as (a) a validation source for the classical code, and (b) a very small external check on 8 unseen cases" — never "benchmarked on the full internationally recognised IEC set" without the overlap disclosure.

OBSERVED IN FILE

ENGINEERING INTERPRETATION

Part 34 — Scientific validity: two very different sentences

Compare: **(A)** "The model scored 95% on our test file." **(B)** "The model can diagnose real-world transformers with 95% accuracy." **ENGINEERING INTERPRETATION**

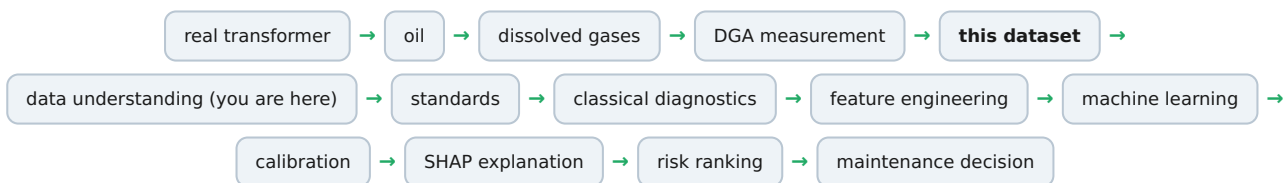
(A) is a measurement on a specific, small, curated file. (B) is a claim about the entire world of transformers. Getting from (A) toward (B) requires: a large enough sample, a realistic class distribution, trustworthy labels, varied measurement conditions, and ideally multiple independent sources. Your data satisfies few of these fully. Therefore you may state (A) with its exact conditions, and you must **not** state (B). The gap between them is where over-claiming projects lose credibility.

Part 35 — What this dataset can and cannot prove

CAN legitimately support	CANNOT prove from this data
Classifying a sample's fault type from its gases	Exact remaining useful life of a transformer
Comparing classical methods vs ML on the same cases	The date/time a transformer will fail
Feature-engineering experiments (ratios, logs)	Guaranteed real-world field accuracy
Building an explainable, calibrated classifier	Universal validity across all transformer types
Demonstrating the full standards+ML pipeline	Continuous online monitoring or trending

State this honestly in your report. "A well-executed classifier on a recognised public DGA dataset, benchmarked against classical methods" is a strong, true claim that earns interviews. Anything about deployment-grade accuracy or remaining life would be false given this data. **ENGINEERING INTERPRETATION**

Part 36 — The dataset's place in the whole project



Every later step in your handbook consumes the understanding you built here. The dataset is the foundation; if you misunderstood it, everything above would inherit the mistake — which is exactly why this stage was worth doing carefully.

ENGINEERING INTERPRETATION

Part 37 — Dataset master comparison table

Item	Training	Unseen Test	IEC TC 10
File name	DGA_train.xlsx	DGA_test_unseen.xlsx	IEC_TC_10_data.xlsx
Rows	584	70	49
Columns	6 (5 gases + Fault class) — identical in all three		
Target classes	7 present	7 present	7 present
Class shape	uneven (45-134)	flat (10 each)	steep (1-15)
Missing values	0	0	0
Exact duplicates	2	0	0
Overlap with Train	—	0 rows	41 of 49
Documented purpose	training / tuning	final generalization test	benchmark reference
Safe recommended use	train + stratified CV	evaluate once, at the end	classical validation + 8-row external check
Key limitation	"balanced" is overstated	curated, not real-world frequencies	84% leaks into Train

Part 38 — Gas dictionary & fault dictionary

38.1 Gas dictionary

Gas	Formula	Simple meaning	Physical origin	Fault relevance	Train range (min-max ppm)
Hydrogen	H ₂	hydrogen	any fault, easiest to form	partial discharge / general	0.0000 – 92600
Methane	CH ₄	methane	mild heating	low-temp thermal	0.0000 – 61000
Ethane	C ₂ H ₆	ethane	low-temp heating	warm thermal	0.0000 – 29990
Ethylene	C ₂ H ₄	ethylene	hot heating	high-temp thermal	0.0000 – 42100
Acetylene	C ₂ H ₂	acetylene	needs most energy	arcing / high-energy discharge	0.0000 – 57000

38.2 Fault dictionary

Code	Fault	Simple meaning	Gas pattern	Train count
0	Normal / no detected fault	no fault detected	all gases low	85
1	Partial Discharge (PD)	gentle sparking	mostly H ₂	59
2	Low-energy discharge (D1)	low-energy sparking	H ₂ +C ₂ H ₂	76
3	High-energy discharge / arcing (D2)	violent arcing	strong C ₂ H ₂	134
4	Thermal fault < 300 C (T1)	mild overheating	CH ₄ /C ₂ H ₆	96
5	Thermal fault 300-700 C (T2)	moderate overheating	C ₂ H ₄ rising	45
6	Thermal fault > 700 C (T3)	severe overheating	strong C ₂ H ₄	89

Part 39 — Index of figures in this document

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The workflow diagrams (measurement chain, without-ML, with-ML, dataset-to-decision) appear inline in Parts 11, 28, 29 and 36.

Part 40 — Question bank (test yourself)

Beginner

1. What is a dataset? a row? a column?
2. What is a transformer and why does it hold oil?
3. What does DGA stand for and what does it measure?
4. What is ppm?
5. Name the five gases in your files.

Intermediate

1. Why does hydrogen appear in almost every fault?
2. Why is acetylene the "alarm" gas?
3. What does each fault code 0-6 mean physically?
4. What is IEC TC 10 and why is a reference set useful?
5. What is IEC 60599 and what does it provide?
6. Why do engineers use gas ratios instead of raw values?

Advanced

1. Why is your training set not truly "balanced," and how did you show it?
2. Why is overall accuracy a misleading metric here?
3. What is data leakage, and exactly where does it appear in your three files?
4. Why might a model do well on DGA_train yet poorly on genuinely new transformers?
5. Why can this data not support remaining-useful-life claims?
6. Why is deep learning the wrong tool at 584 rows?
7. What would you have to know (that is missing) to claim field-grade accuracy?

Part 41 — The whole thing, as simply as possible

We have three boxes of transformer "blood-test" records. Each record lists how much of five gases was found in a transformer's oil. The gases are clues left behind by heat or sparking inside the transformer. A final number says which condition that sample matched. One big box (584 records) teaches the computer; one small sealed box (70) tests whether it really learned; a third box (49) is a set of well-known reference cases for comparison. The catch we discovered: most of that third box is secretly a copy of the teaching box, so it cannot be used as a fair independent test. The clues are real, the data is clean, but it is small and tidied-up — good enough to learn and demonstrate the methods, not to certify real transformers.

Part 42 — Technical summary (for an evaluator)

Structure: three anonymised DGA tables (584 / 70 / 49) with five gas features (H_2 , CH_4 , C_2H_6 , C_2H_4 , C_2H_2 , ppm assumed) and a 7-class integer target following the IEC-style taxonomy (Normal, PD, D1, D2, T1-T3). **Provenance:** IEEE DataPort (DOI 10.21227/27vy-h479); transformer metadata, timestamps, and label methodology undocumented. **Quality:** no missing values; minor exact duplication in Train; right-skewed gas distributions; mixed zero / 0.0001 handling. **Distribution:** Train mildly imbalanced (documentation's "balanced" overstated), Test uniform (curated), IEC steeply imbalanced. **Critical issue:** 41/49 IEC rows overlap Train — precludes naive external benchmarking. **Formulation:** multi-class classification on a small tabular set; classical baselines (Key Gas, IEC 60599 ratios, Duval) first, then tree ensembles / SVM with stratified CV, probability calibration, and SHAP; per-class metrics over accuracy. **External validity:** low without population metadata; snapshot data precludes RUL/trending. **Industrial relevance:** demonstrates a standards-grounded, explainable DGA classification pipeline; not a deployment-grade field tool.

Ownership test — "Can I now explain this dataset?"

Tick each only if you can answer it out loud without looking. Any you cannot is a marked learning gap to revisit.

- ☐ What is each of the three files, and its role?
- ☐ Why are there three files, not one?
- ☐ Where did the data come from, and what is undocumented?
- ☐ What does one row represent (and why is it not "one transformer")?
- ☐ What does every column mean, and its unit?
- ☐ How are the gas values physically measured?
- ☐ What does each of the five gases signal?
- ☐ What does each fault code 0-6 mean?
- ☐ What does "Normal" really mean (and not mean)?
- ☐ What is IEC TC 10 and what are IEEE/IEC standards?

- How would an engineer diagnose without ML?
- Why do we add ML, and what does it actually learn?
- Why is the training set not truly balanced?
- Why is accuracy a dangerous metric here?
- Where is the data leakage, and how will you handle it?
- What are the dataset's limits, and what it can vs cannot prove?
- How does the dataset fit the full project pipeline?

If you can tick all seventeen, you no longer merely "have three Excel files." You understand what they are, where they came from, what every number means, how engineers and machine learning use them, and what they can and cannot prove — which was the whole point. Now the next real step is code: Step 9, `key_gas.py` . **ENGINEERING INTERPRETATION**